UC SAN DIEGO / ECE 251C

3

Finite-energy signals and sampling

Executive summary

The spaces $\ell^2(\mathbb{Z})$ and $L^2(\mathbb{R})$ describe finite-energy sequences and functions. Their inner products connect signal energy to Fourier analysis through Parseval's and Plancherel's identities. Ideal sampling, expressed as multiplication by a Dirac comb, repeats the original spectrum at intervals of $\omega_0 = 2\pi/T$. If these copies overlap, different signals can have the same samples. If a signal is bandlimited to B and $\omega_0 > 2B$, ideal low-pass filtering recovers its spectrum. In the signal domain, this reconstruction is a sum of shifted sinc functions weighted by the samples.

3.1 Finite-energy sequences: the space $\ell^2(\mathbb{Z})$

For a finite vector $x \in \mathbb{C}^N$, the squared Euclidean norm is $\|x\|_2^2 = \sum_{n=0}^{N-1} |x[n]|^2$. We extend this notion to a sequence indexed by all integers by summing the squared magnitudes of all its entries. The resulting quantity is the *energy*:

$$\|x\|_{\ell^2}^2 := \sum_{n\in\mathbb{Z}} |x[n]|^2, \qquad \ell^2(\mathbb{Z}) := \left\{ x : \mathbb{Z} \to \mathbb{C} : \sum_{n\in\mathbb{Z}} |x[n]|^2 < \infty \right\}. \tag{3.1}$$

The norm $\|x\|_{\ell^2}$ is the square root of the energy. We use subscript 2 for the Euclidean norm of a finite vector and subscript ℓ^2 for the norm of a sequence on $\mathbb{Z}$. A finite vector can be viewed as a sequence by extending it by zero; in that case the two norms have the same numerical value. The distinction in notation records the underlying domain.

An infinite sequence can have finite energy. For $x[n] = 2^{-|n|}$,

$$\|x\|_{\ell^2}^2 = 1 + 2\sum_{n=1}^{\infty} 4^{-n} = 1 + 2\frac{1/4}{1-1/4} = \frac{5}{3}.$$

3.2 Inner products of sequences

For real finite vectors, $\langle x, y\rangle = x^{\mathsf{T}} y = \sum_{n=0}^{N-1} x[n]y[n]$. For $x, y \in \ell^2(\mathbb{Z})$, use

$$\langle x, y\rangle_{\ell^2} := \sum_{n\in\mathbb{Z}} x[n]y^*[n], \qquad \|x\|_{\ell^2} = \sqrt{\langle x, x\rangle_{\ell^2}}. \tag{3.2}$$

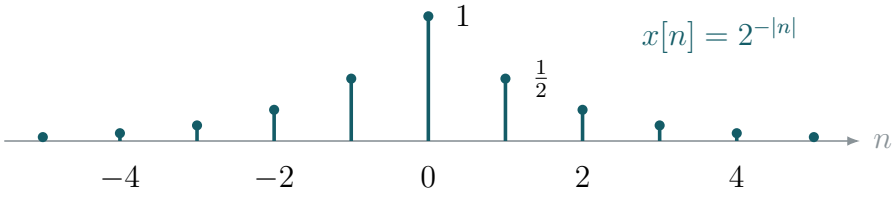


Figure 3.1: Infinitely many nonzero entries, but a finite sum of squared magnitudes. Finite energy is not the same as finite support.

The superscript $*$ denotes complex conjugation: $(a + \mathrm{j}b)^* = a - \mathrm{j}b$ for real a, b. The series in (3.2) is absolutely convergent by the Cauchy–Schwarz inequality. Our convention conjugates the second argument, so the inner product is *linear in its first argument* and conjugate-linear in its second:

$$\langle ax + by, z\rangle = a\langle x, z\rangle + b\langle y, z\rangle, \qquad \langle x, ay\rangle = a^*\langle x, y\rangle.$$

Since $x[n]x^*[n] = |x[n]|^2$, we have $\langle x, x\rangle_{\ell^2} = \|x\|_{\ell^2}^2$. In other words, the inner product *induces the* ℓ^2 *norm*. Without conjugation, a sequence with only $x[0] = \mathrm{j}$ nonzero would give -1 instead of energy one.

3.3 Finite-energy functions: the space $L^2(\mathbb{R})$

For a continuous-domain signal, replace the sum by an integral:

$$\|f\|_{L^2}^2 := \int_{\mathbb{R}} |f(t)|^2 \,\mathrm{d}t, \qquad L^2(\mathbb{R}) := \left\{ f : \mathbb{R} \to \mathbb{C} : \int_{\mathbb{R}} |f(t)|^2 \,\mathrm{d}t < \infty \right\}. \tag{3.3}$$

We work with measurable functions, so these integrals are defined, and identify functions that agree except on a set of length zero. This qualification will matter when we discuss point sampling. The corresponding inner product and the norm it induces are

$$\langle f, g\rangle_{L^2} := \int_{\mathbb{R}} f(t)g^*(t) \,\mathrm{d}t, \qquad \|f\|_{L^2} = \sqrt{\langle f, f\rangle_{L^2}}. \tag{3.4}$$

Setting $g = f$ gives $\langle f, f\rangle_{L^2} = \int_{\mathbb{R}} |f(t)|^2 \,\mathrm{d}t = \|f\|_{L^2}^2$. Thus the inner product induces the energy norm in both domains. The domain distinction remains important: a function on $\mathbb{R}$ is not a sequence on $\mathbb{Z}$, and a continuous-domain signal need not be a continuous function.

Example: a pulse with infinite extent. For $f(t) = \mathrm{e}^{-|t|}$,

$$\|f\|_{L^2}^2 = 2\int_0^\infty \mathrm{e}^{-2t} \,\mathrm{d}t = 2\left[-\tfrac{1}{2}\mathrm{e}^{-2t}\right]_0^\infty = 1.$$

A nonzero constant or a persistent sinusoid has infinite energy on $\mathbb{R}$. These signals remain useful for studying frequency response, but are not in $L^2(\mathbb{R})$.

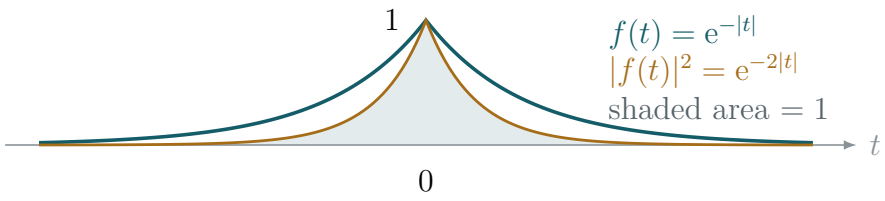


Figure 3.2: Energy is area under the *squared magnitude*, not area under the signal itself.

3.4 Other measures of size: ℓ^p and L^p

Historical remark. The L^p spaces are called *Lebesgue spaces*, after the French mathematician Henri Lebesgue. The corresponding sequence spaces are written ℓ^p. The letters ℓ and L thus provide a useful reminder of this connection between sequence spaces and function spaces.

Energy uses the square of the magnitude, but other powers also give useful ways to measure a signal. For $1 \le p < \infty$, the corresponding definitions are

$$\|x\|_{\ell^p} := \left(\sum_{n \in \mathbb{Z}} |x[n]|^p \right)^{1/p}, \qquad \|f\|_{L^p} := \left(\int_{\mathbb{R}} |f(t)|^p \, \mathrm{d}t \right)^{1/p}. \tag{3.5}$$

The spaces $\ell^p(\mathbb{Z})$ and $L^p(\mathbb{R})$ consist of signals for which the respective expression is finite. The choice $p = 1$ measures absolute summability or integrability, whereas $p = 2$ measures the square root of energy. These different ways of measuring size will lead to different penalties when we study signal recovery. In particular, absolute-value penalties will be central to sparsity.

At $p = \infty$, use the smallest uniform bound:

$$\|x\|_{\ell^\infty} = \sup_{n \in \mathbb{Z}} |x[n]|, \qquad \|f\|_{L^\infty} = \operatorname*{ess\,sup}_{t \in \mathbb{R}} |f(t)|.$$

A *supremum* is the least upper bound. It need not be an attained maximum: the sequence $x[n] = 1 - 1/(n+1)$ for $n \ge 0$, extended by zero for $n < 0$, has supremum one but no entry equal to one.

Functions that differ only on a set of total length zero agree *almost everywhere* and represent the same element of L^p. An *essential supremum* is the smallest bound allowed to fail on such a set. For example, let $g(t) = 1$ on $[-1, 1]$, set $g(0) = 3$, and set $g(t) = 0$ outside the interval. Then

$$\sup_t |g(t)| = 3, \qquad \|g\|_{L^\infty} = 1, \qquad \|g\|_{L^2}^2 = 2.$$

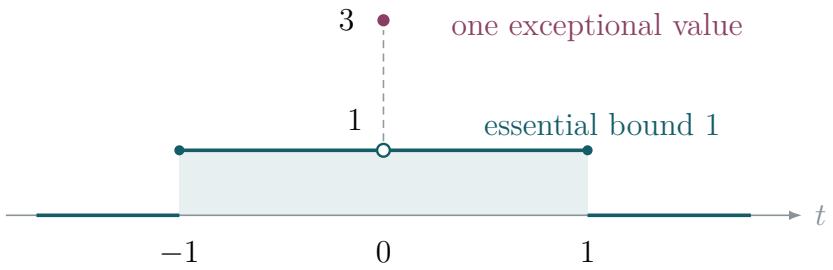


Figure 3.3: An isolated value changes the ordinary supremum, but not an integral or the essential supremum. The dashed guide is not part of the signal.

> **Why this matters for sampling**
> Energy alone does not specify point values. In ideal sampling we take f to be continuous, so the sample values $f[n] = f(nT)$ are unambiguous.

3.5 Parseval and Plancherel for sequences

Fourier analysis preserves inner products and energy, with constants determined by the transform convention. For sequences we write

$$\widehat{x}(\theta) = X(\mathrm{e}^{\mathrm{j}\theta}) = \sum_{n\in\mathbb{Z}} x[n]\mathrm{e}^{-\mathrm{j}n\theta}, \qquad -\pi \le \theta \le \pi. \tag{3.6}$$

We use θ for the *discrete-domain* angular frequency. The spectrum repeats every 2π because $\mathrm{e}^{-\mathrm{j}n(\theta+2\pi)} = \mathrm{e}^{-\mathrm{j}n\theta}$ for integer n.

The calculation for finite sequences. For integers m, n,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{e}^{-\mathrm{j}(n-m)\theta}\,\mathrm{d}\theta = \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases}$$

For $n = m$ we integrate one. Otherwise the antiderivative is $\mathrm{e}^{-\mathrm{j}(n-m)\theta}/[-\mathrm{j}(n-m)]$, with equal values at both endpoints. Therefore

$$\begin{aligned} \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{x}(\theta)\widehat{y}^*(\theta)\,\mathrm{d}\theta &= \sum_n \sum_m x[n]y^*[m]\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{e}^{-\mathrm{j}(n-m)\theta}\,\mathrm{d}\theta \\ &= \sum_n x[n]y^*[n] = \langle x, y\rangle_{\ell^2}. \end{aligned} \tag{3.7}$$

All sums are finite here. Setting $y = x$ proves the energy identity. The same integral identity extracts one coefficient:

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{x}(\theta)\mathrm{e}^{\mathrm{j}n\theta}\,\mathrm{d}\theta. \tag{3.8}$$

The finite-energy identities. Parseval and Plancherel extend the displayed identities to all $x, y \in \ell^2(\mathbb{Z})$. In particular,

$$\boxed{\|x\|_{\ell^2}^2 = \frac{1}{2\pi}\|\widehat{x}\|_{L^2[-\pi,\pi]}^2.} \tag{3.9}$$

For a general ℓ^2 sequence, the series defining the DTFT is interpreted as a limit in $L^2[-\pi, \pi]$; pointwise or absolute convergence is not required. Conversely, an L^2 function on $[-\pi, \pi]$ defines an ℓ^2 sequence through the inverse formula (3.8). Thus finite energy of the sequence is equivalent to finite energy of its spectrum over one period. Equation (3.9) identifies the energy of the sequence with the normalized spectral energy over *one period*. Integrating a nonzero periodic spectrum over the entire real line would instead give infinite energy.

3.6 The continuous-domain energy identity

Our angular-frequency convention is

$$\widehat{f}(\omega) = \int_{\mathbb{R}} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \qquad f(t) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega. \tag{3.10}$$

Theorem 3.1 (Parseval and Plancherel)**.** The Fourier transform extends to finite-energy signals. For $f, g \in L^2(\mathbb{R})$,

$$\langle f, g\rangle_{L^2(\mathbb{R})} = \frac{1}{2\pi}\langle \widehat{f}, \widehat{g}\rangle_{L^2(\mathbb{R})}, \tag{3.11}$$

$$\|f\|_{L^2}^2 = \frac{1}{2\pi}\|\widehat{f}\|_{L^2}^2. \tag{3.12}$$

Every L^2 spectrum is the transform of a unique L^2 signal, with equality of functions understood almost everywhere.

Where the identity comes from. For $f, \widehat{g} \in L^1(\mathbb{R})$ with Fourier inversion valid for g, substitute the inverse transform into the inner product:

$$\begin{aligned}\int_{\mathbb{R}} f(t)g^*(t)\,\mathrm{d}t &= \int_{\mathbb{R}} f(t)\left[\frac{1}{2\pi}\int_{\mathbb{R}} \widehat{g}^*(\omega)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}\omega\right]\mathrm{d}t \\ &= \frac{1}{2\pi}\int_{\mathbb{R}}\left[\int_{\mathbb{R}} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t\right]\widehat{g}^*(\omega)\,\mathrm{d}\omega \\ &= \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\omega)\widehat{g}^*(\omega)\,\mathrm{d}\omega.\end{aligned}$$

This calculation gives the inner-product identity whenever the stated integrability conditions justify the interchange. Taking $g = f$ gives the energy identity. The

general theorem extends the transform and these identities to all finite-energy signals by L^2 approximation. Thus the Fourier transform remains meaningful even when its defining integral is not absolutely convergent.

Check the normalization. For $x[n] = \delta[n] + \delta[n-1]$, the DTFT is $1 + \mathrm{e}^{-\mathrm{j}\theta}$, so

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |1 + \mathrm{e}^{-\mathrm{j}\theta}|^2 \,\mathrm{d}\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} (2 + 2\cos\theta)\,\mathrm{d}\theta = 2 = \|x\|_{\ell^2}^2.$$

For the unit box from Lecture 2, $\|h\|_{L^2}^2 = 1$ and $\widehat{h}(\omega) = \mathrm{sinc}(\omega/(2\pi))$, where $\mathrm{sinc}(u) = \sin(\pi u)/(\pi u)$ with value one at zero. Plancherel and $u = \omega/(2\pi)$ give

$$1 = \frac{1}{2\pi}\int_{\mathbb{R}} \left|\mathrm{sinc}\left(\frac{\omega}{2\pi}\right)\right|^2 \mathrm{d}\omega = \int_{\mathbb{R}} |\,\mathrm{sinc}(u)|^2 \,\mathrm{d}u.$$

3.7 Ideal sampling: multiplication by a Dirac comb

We now use the continuous and discrete descriptions together. Let f be continuous and let $T > 0$ be the sampling step. Sampling records the sequence $f[n] \coloneqq f(nT)$. Continuity specifies the point values that an energy-space description alone does not determine. Ideal sampling records those values at the locations nT, $n \in \mathbb{Z}$. To express the operation using continuous-domain notation, define the *Dirac comb*

$$\text{Ш}_T(t) \coloneqq \sum_{n\in\mathbb{Z}} \delta(t - nT). \tag{3.13}$$

It is a train of unit-weight impulses, spaced T apart. On any bounded interval there are only finitely many impulses. Ideal sampling multiplies the input by this comb. By the sifting property,

$$\begin{aligned} f_T(t) \coloneqq f(t)\text{Ш}_T(t) &= \sum_{n\in\mathbb{Z}} f(t)\delta(t - nT) \\ &= \sum_{n\in\mathbb{Z}} f[n]\delta(t - nT), \qquad f[n] = f(nT). \end{aligned} \tag{3.14}$$

The value $f(nT)$ is the *weight* of the impulse at nT.

The *sample sequence* $f[n]$ lives on $\mathbb{Z}$, while the impulse train f_T lives on $\mathbb{R}$. They encode the same numbers in different mathematical objects. A Dirac comb or a nonzero sampled impulse train is not an ordinary L^2 function: we do not square impulses to assign it an energy.

3.8 The spectrum of the sampled signal is periodic

We first carry out the Fourier calculation for a smooth signal that is zero outside a bounded interval, so that the sampled sum has only finitely many nonzero terms.

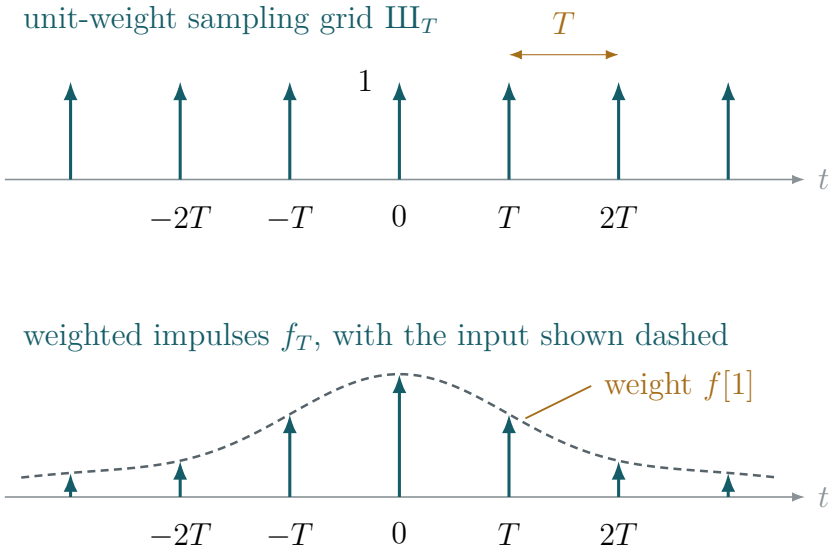


Figure 3.4: Multiplication by the comb retains the values on the grid. The input between samples is not recorded. The impulse train is not a staircase or a line joining the samples.

The finite-energy bandlimited interpretation used for reconstruction is explained below. Applying the Fourier transform to equation (3.14) gives

$$\begin{aligned}\widehat{f}_T(\omega) &= \int_{\mathbb{R}} \sum_{n\in\mathbb{Z}} f[n]\delta(t-nT)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \\ &= \sum_{n\in\mathbb{Z}} f[n] \int_{\mathbb{R}} \delta(t-nT)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \\ &= \sum_{n\in\mathbb{Z}} f[n]\mathrm{e}^{-\mathrm{j}\omega nT}. \end{aligned} \tag{3.15}$$

The sampling angular frequency is $\omega_0 := 2\pi/T$. Since n is an integer,

$$\mathrm{e}^{-\mathrm{j}(\omega+\omega_0)nT} = \mathrm{e}^{-\mathrm{j}\omega nT}\mathrm{e}^{-\mathrm{j}2\pi n} = \mathrm{e}^{-\mathrm{j}\omega nT}.$$

Consequently, $\widehat{f}_T(\omega+\omega_0) = \widehat{f}_T(\omega)$: the spectrum repeats every $2\pi/T$.

Physical and normalized frequency. The sampled sequence has DTFT

$$F(\mathrm{e}^{\mathrm{j}\theta}) = \sum_{n\in\mathbb{Z}} f[n]\mathrm{e}^{-\mathrm{j}n\theta}.$$

The DTFT $F(\mathrm{e}^{\mathrm{j}\theta})$ describes the sample sequence, whereas $\widehat{f}(\omega)$ describes the original function. The transform of the impulse train uses the same sample values as the DTFT, but a different frequency coordinate. Comparing the exponents in the two formulas gives

$$\boxed{\widehat{f}_T(\omega) = F(\mathrm{e}^{\mathrm{j}\omega T}), \qquad \theta = \omega T.} \tag{3.16}$$

The variable ω measures radians per unit of the continuous coordinate; θ measures radians per sample. They coincide only when $T = 1$.

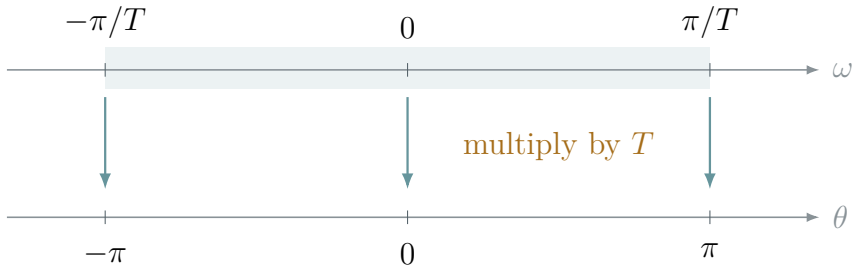


Figure 3.5: The base interval $[-\pi/T, \pi/T]$ has width $2\pi/T$. In the normalized variable $\theta = \omega T$, it becomes $[-\pi, \pi]$.

A nonzero periodic spectrum has infinite energy when integrated over *all* of $\mathbb{R}$. For a sequence we integrate over just one period, as in equation (3.9). There is no conflict with the continuous-domain theorem: the sampled object f_T is an impulse train, not an $L^2(\mathbb{R})$ function.

Periodicity tells us *where* the spectrum repeats. To determine *what* is repeated, we next compute the Fourier transform of the sampling comb.

3.9 The Fourier transform of a Dirac comb

Recall the Fourier series of a T-periodic signal g:

$$g(t) = \sum_{k\in\mathbb{Z}} c_k \mathrm{e}^{\mathrm{j}k\omega_0 t}, \qquad c_k = \frac{1}{T}\int_{-T/2}^{T/2} g(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t.$$

The comb has one impulse, at zero, in this central period. Its coefficients are therefore

$$c_k = \frac{1}{T}\int_{-T/2}^{T/2} \mathrm{Ш}_T(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t = \frac{1}{T}.$$

Hence $\mathrm{Ш}_T(t) = T^{-1}\sum_k \mathrm{e}^{\mathrm{j}k\omega_0 t}$. This is an *impulse identity*: as with the Dirac impulse, it is understood through integration.

Transforming one exponential. The inverse transform of $2\pi\delta(\omega - a)$ is

$$\frac{1}{2\pi}\int_{\mathbb{R}} 2\pi\delta(\omega - a)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega = \mathrm{e}^{\mathrm{j}at}.$$

Thus $\mathrm{e}^{\mathrm{j}at} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega - a)$. This exchange is related to *Fourier duality*: whenever the ordinary integrals and inversion hold, inversion at $-\omega$ gives

$$f(t) \overset{\mathcal{F}}{\longleftrightarrow} \widehat{f}(\omega) \quad \Longrightarrow \quad \widehat{f}(t) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi f(-\omega).$$

Now transform the comb's Fourier series term by term as an impulse identity:

$$\boxed{\widehat{\mathrm{III}}_T(\omega) = \frac{2\pi}{T} \sum_{k\in\mathbb{Z}} \delta\!\left(\omega - \frac{2\pi k}{T}\right).} \tag{3.17}$$

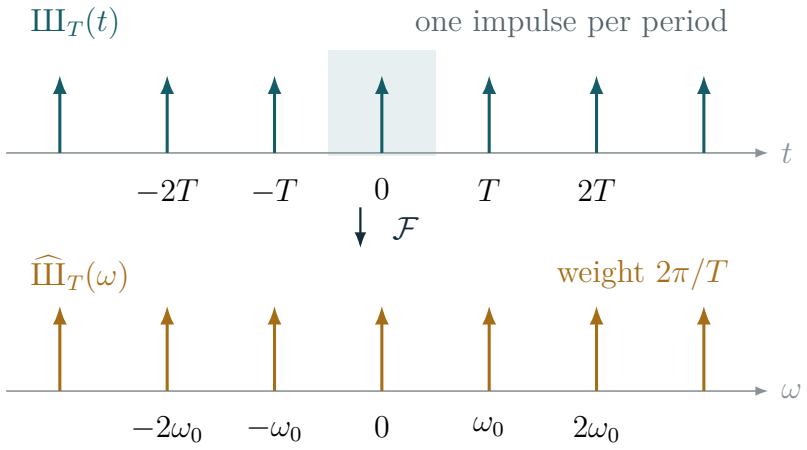


Figure 3.6: A comb transforms into a comb. Spatial spacing T becomes frequency spacing $2\pi/T$. The shaded interval on top is one sampling period; the arrows below have weight $2\pi/T$, not weight one.

3.10 Sampling periodizes the original spectrum

Multiplication in the signal domain becomes convolution in the Fourier domain, with our factor $1/(2\pi)$. For $f, \hat{g} \in L^1(\mathbb{R})$ with Fourier inversion valid for g, substitute the inverse transform of g to derive

$$\begin{aligned}\widehat{fg}(\omega) &= \int_{\mathbb{R}} f(t) \left[\frac{1}{2\pi}\int_{\mathbb{R}} \hat{g}(\nu) \mathrm{e}^{\mathrm{j}\nu t}\,\mathrm{d}\nu\right] \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \\ &= \frac{1}{2\pi}\int_{\mathbb{R}} \hat{g}(\nu)\hat{f}(\omega-\nu)\,\mathrm{d}\nu = \frac{1}{2\pi}(\hat{f} * \hat{g})(\omega).\end{aligned}$$

Apply the corresponding impulse identity with $g = \mathrm{III}_T$. Convolution with a shifted impulse translates the spectrum, so equation (3.17) gives

$$\begin{aligned}\hat{f}_T(\omega) &= \frac{1}{2\pi}\left(\hat{f} * \frac{2\pi}{T}\sum_k \delta(\cdot - k\omega_0)\right)(\omega) \\ &= \frac{1}{T}\sum_{k\in\mathbb{Z}} \hat{f}(\omega - k\omega_0).\end{aligned} \tag{3.18}$$

Consequently, the spectrum of the sampled signal is obtained by translating the original spectrum by integer multiples of the sampling angular frequency

and adding the copies with weight $1/T$. Combining the two expressions for this spectrum yields the *Poisson summation formula*:

$$\boxed{\sum_{n\in\mathbb{Z}} f[n]\mathrm{e}^{-\mathrm{j}\omega nT} = \frac{1}{T}\sum_{k\in\mathbb{Z}} \widehat{f}\left(\omega - \frac{2\pi k}{T}\right).} \tag{3.19}$$

For a finite-energy bandlimited signal, the right side is an L^2 function on one period. Its nth Fourier-series coefficient is $f(nT)$, by the inverse Fourier formula. Thus (3.19) also holds in L^2 on a period for this class; neither side needs an absolutely convergent pointwise series. This is the interpretation used in the reconstruction argument.

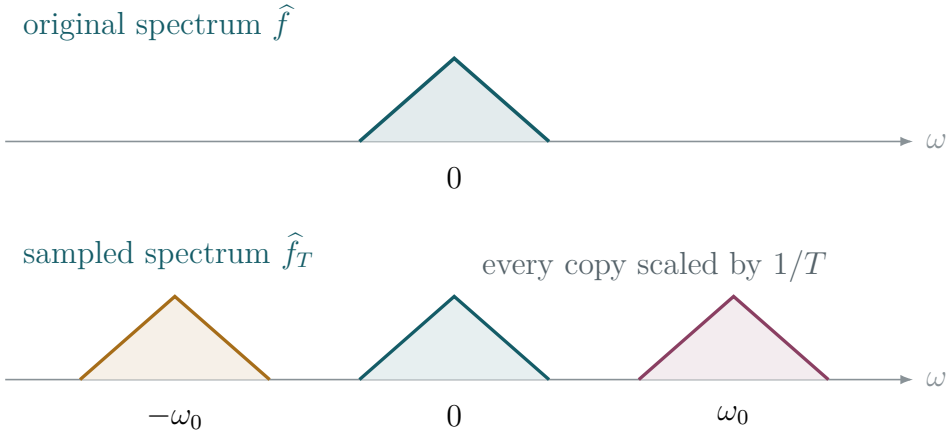


Figure 3.7: Sampling translates copies of the original spectrum by multiples of $\omega_0 = 2\pi/T$ and adds them. The vertical scales of the two rows differ by $1/T$. The triangle is an illustrative spectrum.

3.11 Aliasing is spectral overlap

Suppose $\widehat{f}(\omega) = 0$ for $|\omega| > B$. We call f *bandlimited* to B. Its translated spectral copies have width $2B$ and their centers are spaced $\omega_0 = 2\pi/T$ apart.

If $\omega_0 > 2B$, neighboring copies are separated. If $\omega_0 < 2B$, they can overlap. In an overlap region, the observed value is a *sum* of different spectral contributions. We generally cannot separate those contributions from their sum.

Increasing the sample spacing T decreases the separation $\omega_0 = 2\pi/T$ between spectral copies. Once distinct copies overlap, the measured spectrum contains their sum rather than the individual contributions. A sufficient condition that prevents this overlap for a signal occupying the entire band is

$$\boxed{\omega_0 > 2B \quad\Longleftrightarrow\quad T < \frac{\pi}{B}.} \tag{3.20}$$

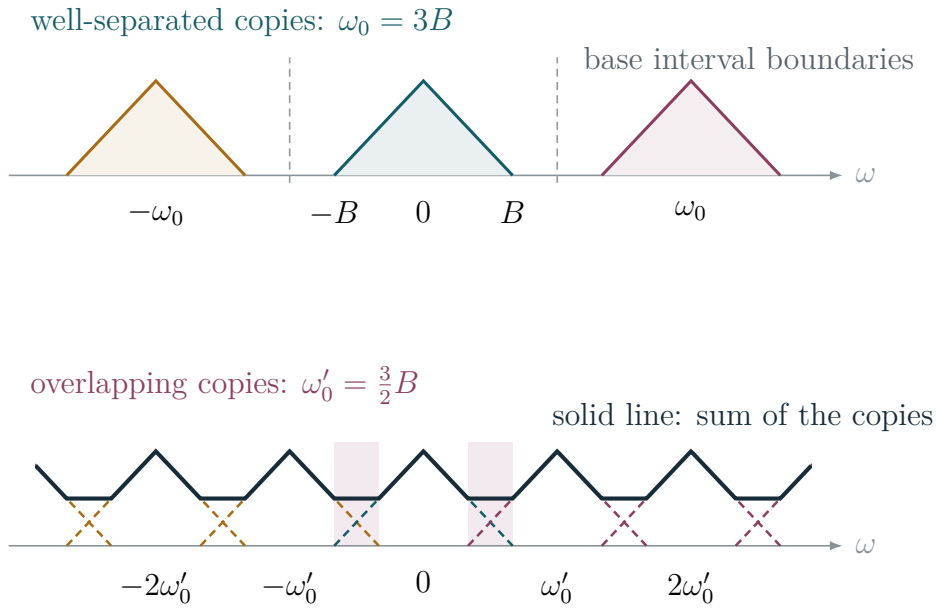


Figure 3.8: The same triangular spectrum sampled at two different steps. To compare the shapes, the rows show $T\widehat{f}_T$ and $T'\widehat{f}_{T'}$, respectively. Dashed curves in the lower row are individual copies; the solid curve is their sum. Shading marks overlap with the central copy.

If frequency is measured in cycles per unit, write $\widetilde{B} = B/(2\pi)$ and $\nu_0 = 1/T$. The same inequality becomes $\nu_0 > 2\widetilde{B}$. The strict inequality leaves a gap between the copies.

3.12 Different functions can have exactly the same samples

Spectral overlap has a concrete consequence: the samples need not distinguish two different signals. Here is an example *within the finite-energy setting.* Let

$$q(t) = \operatorname{sinc}\left(\frac{t}{4T}\right), \qquad u(t) = q(t)\cos\left(\frac{2\pi t}{5T}\right), \qquad v(t) = q(t)\cos\left(\frac{8\pi t}{5T}\right).$$

At every sampling location, the cosine factors agree:

$$\cos\left(\frac{8\pi n}{5}\right) = \cos\left(2\pi n - \frac{2\pi n}{5}\right) = \cos\left(\frac{2\pi n}{5}\right).$$

Consequently $u(nT) = v(nT)$ for every integer n. But at $t = T/2$, the two cosine factors are $\cos(\pi/5)$ and $\cos(4\pi/5) = -\cos(\pi/5)$; since $q(T/2) \neq 0$, the functions are different.

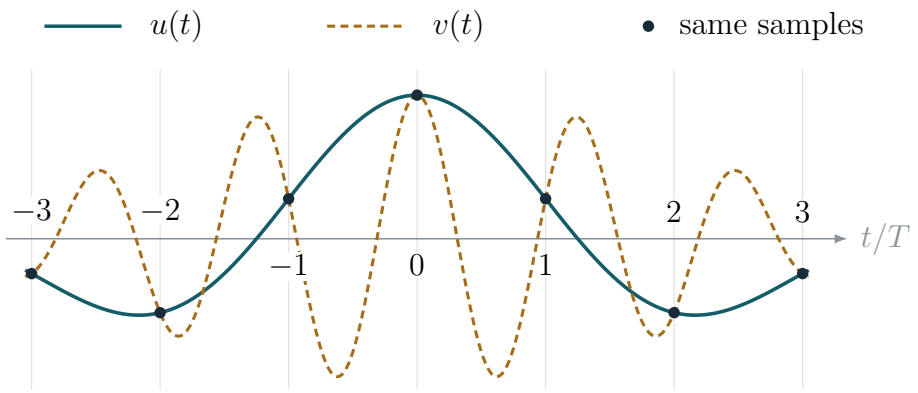


Figure 3.9: Two continuous functions pass through the same sample values but differ between samples. The solid and dashed curves are u and v; the dots are their common samples.

These are not infinite-energy pure tones. Using $|\cos| \le 1$ and the sinc energy computed earlier,

$$\|u\|_{L^2}^2,\ \|v\|_{L^2}^2 \le \int_{\mathbb{R}} \left|\operatorname{sinc}\left(\frac{t}{4T}\right)\right|^2 \mathrm{d}t = 4T \int_{\mathbb{R}} |\operatorname{sinc}(s)|^2 \,\mathrm{d}s = 4T.$$

Sampling requires a signal model

A recovery theorem cannot hold for all finite-energy functions. A bandwidth restriction rules out ambiguous alternatives. The theorem will recover the function in the specified bandlimited class, not every function that happens to pass through the samples.

3.13 Recovering the spectrum by ideal low-pass filtering

Assume $B < \pi/T$. On the base interval $[-\pi/T, \pi/T]$, the only possible nonzero term in equation (3.18) is the central copy. Thus

$$\widehat{f}_T(\omega) = \frac{1}{T}\widehat{f}(\omega) \qquad (|\omega| \le \pi/T),$$

apart from immaterial endpoint conventions. To extract this copy *and restore its scale*, use the response

$$H_T(\omega) := T\mathbf{1}_{[-\pi/T,\pi/T]}(\omega), \qquad H_T(\omega)\widehat{f}_T(\omega) = \widehat{f}(\omega). \tag{3.21}$$

The gain is T, not one: sampling introduced a factor $1/T$. Outside the base interval the reconstruction filter removes every repeated copy.

The bandwidth assumption ensures both that the spectral copies do not overlap and that the reconstruction passband contains the entire original spectrum. The

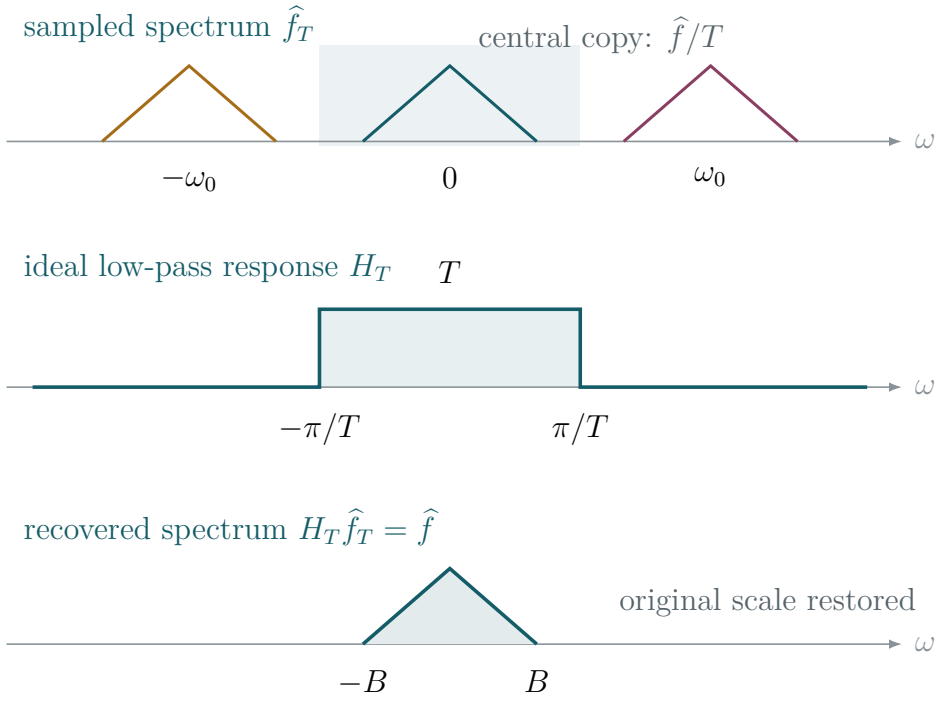


Figure 3.10: Keep one copy, remove the others, and undo the factor $1/T$. The shaded interval is the reconstruction passband. The rows use their own vertical scales.

first condition is essential: filtering can remove unwanted copies outside the passband, but it cannot separate contributions that have already been added at the same frequency inside it.

3.14 The reconstruction kernel is a sinc

Let h_T be the impulse response with Fourier transform H_T from equation (3.21). Invert the rectangle directly. For $t \neq 0$,

$$\begin{aligned} h_T(t) &= \frac{T}{2\pi}\int_{-\pi/T}^{\pi/T} \mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega \\ &= \frac{T}{2\pi}\,\frac{\mathrm{e}^{\mathrm{j}\pi t/T} - \mathrm{e}^{-\mathrm{j}\pi t/T}}{\mathrm{j}t} = \frac{T\sin(\pi t/T)}{\pi t} = \operatorname{sinc}\!\left(\frac{t}{T}\right). \end{aligned} \tag{3.22}$$

At $t = 0$, the same integral is $(T/(2\pi))(2\pi/T) = 1$. Thus the formula includes zero when we define

$$\operatorname{sinc}(s) \coloneqq \begin{cases} \sin(\pi s)/(\pi s), & s \neq 0, \\ 1, & s = 0. \end{cases}$$

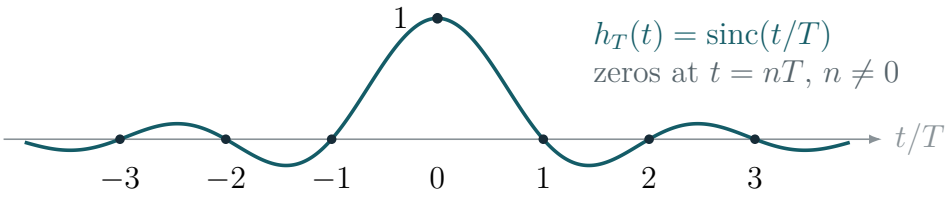


Figure 3.11: The reconstruction kernel is one at its own grid point and zero at every other grid point. The black dots mark its values on the sampling grid.

Filter the sampled impulse train. Each weighted impulse produces a shifted copy of h_T:

$$\begin{aligned}(h_T * f_T)(t) &= \sum_{n\in\mathbb{Z}} f[n] \int_{\mathbb{R}} h_T(t-u)\delta(u-nT)\,\mathrm{d}u \\ &= \sum_{n\in\mathbb{Z}} f[n] h_T(t-nT) \\ &= \sum_{n\in\mathbb{Z}} f[n]\,\mathrm{sinc}\left(\frac{t-nT}{T}\right). \end{aligned} \tag{3.23}$$

At a grid point $t = mT$, the cardinal property of the sinc makes every term except $n = m$ vanish. The remaining term equals $f[m] = f(mT)$, so the expansion interpolates the samples. Interpolation alone would not establish recovery between samples. That stronger conclusion follows from the spectral argument: for a finite-energy signal with the stated bandlimit, the reconstructed spectrum equals $\widehat{f}$. The equality of the resulting continuous functions then gives recovery at every t.

The convergence can also be read from this spectral construction. The finite partial sums in (3.23) have spectra supported on $[-\pi/T, \pi/T]$, and these spectra converge to $\widehat{f}$ in L^2 on that interval. Cauchy–Schwarz then bounds the inverse transform of their difference by its L^2 norm times the square root of the interval length, with the Fourier factor $1/(2\pi)$. This bound is independent of t, so the reconstructed partial sums converge uniformly to the continuous representative of f. The formula therefore recovers the signal between the sample locations as well as at them.

The bandwidth assumption is what makes this recovery possible. Once it is fixed, we can also ask what happens when we sample more densely than necessary or when the recorded values contain noise. These are the questions we take up next.